

Exam

June 14, 2025

Solution hints

1. The term “partial equilibrium analysis” describes the following approach:
 - (a) all market participants are assumed to be price takers.
 - (b) all firms are assumed to have the same production technology.
 - (c) all consumers are assumed to have the same preferences
 - (d) an exchange economy is used to describe a complicated situation of interacting markets in a simplified way.
 - (e) a quasi-linear economy is used as a simplifying model in order to partially capture a complicated situation of interacting markets.

Correct answers: (e)

2. Which of the income streams listed in the answers would a consumer with monotonic preferences choose in a perfect capital market with an interest rate of $r = 50\%$?
 - (a) $(3, 2, 6)$
 - (b) $(3, 3, 6)$
 - (c) $(5, 2, 2)$
 - (d) $(2, 5, 2)$
 - (e) $(2, 2, 5)$

For each income stream (Y_0, Y_1, Y_2) , we can calculate the present value multiplied by $4(1+r)^2$, as $4(1+r)^2Y_0 + 4(1+r)Y_1 + 4Y_2 = 9Y_0 + 6Y_1 + 4Y_2$. For $(Y_0, Y_1, Y_2) = (5, 2, 2)$, we obtain $45 + 12 + 8 = 65$. For $(Y_0, Y_1, Y_2) = (3, 3, 6)$, we obtain $27 + 18 + 24 = 69$. The income streams $(2, 5, 2)$ and $(2, 2, 5)$ are worse than $(5, 2, 2)$ because the high income is achieved later. Furthermore, $(3, 2, 6)$ is of course worse than $(3, 3, 6)$.

Correct answers: (b)

3. A firm uses a single input good and has the cost function $C(q) = q + \sqrt{q}$. Then the returns to scale of the production function are
 - (a) increasing.
 - (b) first increasing and then constant.
 - (c) first increasing and then decreasing.
 - (d) constant.
 - (e) decreasing.

Due to the falling marginal costs $MC(q) = 1 + 1/(2\sqrt{q})$, the production function (as the stretched inverse of the cost function) is convexly curved and thus the returns to scale are increasing.

Correct answers: (a)

4. A consumer's preferences regarding consumption flows (c_0, c_1) for the periods 0 and 1 are represented by the utility function $u(c_0, c_1) = \min\{c_0, 2c_1\}$. The consumer's initial endowment is the income stream $(Y_0, Y_1) = (2, 0)$. The consumer can save money at an interest rate of $r = 10\%$. The following consumption quantity is optimal in period 1:

- (a) 13/10
- (b) 11/21
- (c) 16/11
- (d) 11/16
- (e) 26/11

For the optimal bundle, $c_0^* = 2c_1^*$ and $c_0^* + c_1^*/(1+r) = 2$. Therefore, $2c_1^* + c_1^*/(1+r) = 2$. Furthermore, $1/(1+r) = 1/(11/10) = 10/11$. Therefore, $(32/11)c_1^* = 2$. Therefore, $c_1^* = 11/16$.

Correct answers: (d)

5. Consider a quasi-linear economy in which the market supply, $S(p)$, has a constant price elasticity of 2 and the market demand is equal to $D(p) = 1/p$. In equilibrium, the market price $p^* = 5$ arises. Then the market supply is given by (for all $p > 0$)

- (a) $S(p) = p^{-2}/125$.
- (b) $S(p) = 125p^{-2}$.
- (c) $S(p) = 125p^2$.
- (d) $S(p) = p^2/125$.
- (e) a function different from those given in the other answers.

The equilibrium quantity is $D(p^*) = 1/5$. So $1/5 = S(p^*) = A_S(p^*)^2 = A_S 25$ (see Chapter 9A). So $A_S = 1/125$.

Correct answers: (d)

6. Consider a quasi-linear economy. If all individual supply curves are elastic at an output price p , then the market supply at price p is

- (a) inelastic.
- (b) unit elastic.
- (c) perfectly elastic.
- (d) also elastic.
- (e) perfectly inelastic.

See problem set 10.

Correct answers: (d)

7. A consumer's preferences regarding bundles of goods (x_1, x_2) (with $x_i \geq 0$ for $i = 1, 2$) are represented by the utility function $u(x_1, x_2) = (x_1)^c (x_2)^d$, where $c > 0$ and $d > 0$. The bundle $(1, \sqrt{32})$ is in the upper-contour set of the bundle $(2, 2)$. Then the following condition holds true:

- (a) $c/d \leq 3/2$
- (b) $c/d = 2/3$
- (c) $c/d \geq 2/3$
- (d) $c/d \geq 3/2$
- (e) $c/d = 3/2$

According to the assumptions, $2^{c+d} \leq (\sqrt{32})^d$. Raising both sides by $1/d$ yields $2^{c/d+1} \leq (\sqrt{32})$. Taking the logarithm gives $(c/d + 1) \ln 2 \leq \ln \sqrt{32}$. Because $\sqrt{32} = \sqrt{2^5} = 2^{5/2}$, $\ln \sqrt{32} = (5/2) \ln(2)$. So $(c/d + 1) \ln 2 \leq (5/2) \ln(2)$. Therefore, $(c/d + 1) \leq (5/2)$. Therefore, $c/d \leq 3/2$.

Correct answers: (a)

8. Consider an Edgeworth exchange economy in which each A -trader has the initial endowment $e^A = (5, 12)$ and perfect complement preferences, and each B -trader has the initial endowment $e^B = (12, 5)$ and Cobb-Douglas preferences. Then the set of Pareto-efficient allocations in the Edgeworth box is represented by

- (a) the center of the box.
- (b) the main diagonal in the box (i.e., slope 1).
- (c) the lower edge of the box.
- (d) the upper edge of the box.
- (e) the opposite-to-main diagonal in the box (i.e., slope -1).

The box is a square with side length $5 + 12 = 17$. Given any indifference curve of the B -handlers, the A -handlers are always best placed along the main diagonal because of the L -shaped indifference curves. Every point on the main diagonal in the box belongs to one of the B -indifference curves.

Correct answers: (b)

9. The preferences of a consumer for a bundle of goods (x_1, x_2) (with $x_i \geq 0$ for $i = 1, 2$) are represented by the utility function $u(x_1, x_2) = x_1 - (x_2)^2$. Then the following is *not* true:

- (a) The consumer will consume a bundle whose market value is equal to her income.
- (b) Walras' law.
- (c) The preference relation is monotonic.
- (d) The upper contour sets are convex sets.
- (e) The preference relation is convex.

Monotonicity is violated because utility strictly decreases in x_2 . Since utility strictly increases in x_1 , the consumer will in any case consume a bundle whose market value is equal to her income Y she will consume the bundle $(0, Y/p_1)$. Therefore, Walras' law also applies. The indifference curves are described by equations of the form $\underline{u} = x_1 - (x_2)^2$ (from which it follows in particular that $\underline{u} \leq x_1$). We can also write the equations in the form $x_2 = \sqrt{x_1 - \underline{u}}$. This is the graph of a root function that starts at $x_1 = \underline{u}$. The upper contour (better) sets lie between the graph and the x_1 axis, so they are convex. Therefore, the preference relation is also convex.

Correct answers: (c)

10. A firm that uses a single input good, good 1, has the cost function $C(q) = \sqrt{q} + 3$ for all $q > 0$ and $C(0) = 0$. The input price is $p_1 = 1/2$. Then the production function f has the following property:

- (a) $f(x_1) = (x_1)^2/4 - 6$ for all $x_1 > 6$.
- (b) $f(x_1) = (x_1)^2 - 6$ for all $x_1 > 6$.
- (c) $f(x_1) = (x_1 - 6)^2$ for all $x_1 > 6$.
- (d) $f(x_1) = (x_1 - 6)^2/4$ for all $x_1 > 6$.
- (e) None of the properties specified in the other answers.

How many input units x_1 are needed to produce $q > 0$ units of output? The following applies: $p_1 x_1 = C(q)$, therefore $x_1/2 = \sqrt{q} + 3$, therefore $x_1 = 2\sqrt{q} + 6$, therefore $x_1 > 6$ and $x_1 - 6 = 2\sqrt{q}$, therefore $(x_1 - 6)^2 = 4q$, therefore $(x_1 - 6)^2/4 = q$.

Correct answers: (d)

11. You observe a firm with a perfect-complements production function f that uses the cost-minimizing input combination $(2, 5)$ at the input prices $p_1 = 7$ and $p_2 = 8$. The production function f could then be given as follows: $f(x_1, x_2) =$

- (a) $2x_1 + 5x_2$.
- (b) $\min\{2x_1, 5x_2\}$.
- (c) $\min\{5x_1, 2x_2\}$.
- (d) $\max\{5x_1, 2x_2\}$.
- (e) $\max\{2x_1, 5x_2\}$.

See lecture Chapter 7. The prices are not important here; what matters is that the company demands the two inputs in a ratio of 2 to 5.

Correct answers: (c)

12. A consumer has the utility function $u(x_1, x_2) = \sqrt{x_1} + x_2$ over bundles of goods (x_1, x_2) , where $x_1 \geq 0$ and $x_2 \in \mathbb{R}$. The Engel curves of the consumer in the (x_1, x_2) -diagram (with x_1 on the horizontal axis) then run

- (a) parallel to the main diagonal.
- (b) horizontally.
- (c) parallel to diagonals with slope -1 .
- (d) like a parabola.
- (e) vertically.

Quasilinear preferences, see Chapter 3A.

Correct answers: (e)

13. The marginal product of input 1 in a production function $f(x_1, x_2)$ is defined as

- (a) $\partial f(x_1, x_2) / \partial x_2$.
- (b) $\partial f(x_1, x_2) / \partial x_1$.
- (c) $f(x_1, x_2) / x_2$.
- (d) $f(x_1, x_2) / x_1$.
- (e) $f(x_1, x_2) / (x_1 + x_2)$.

Correct answers: (b)

14. Consider a quasi-linear economy. Initially, the economy is in an equilibrium with price $p^* = 150$ EUR and quantity $q^* = 1000$. At the equilibrium point, the price elasticity of market demand is equal to -3 and the price elasticity of market supply is equal to 1. The government now introduces a small unit tax of $t = 1$ EUR. By approximately how much will the consumer price increase in equilibrium?

- (a) $1/2$
- (b) 1
- (c) $1/3$
- (d) $1/5$
- (e) $1/4$

The marginal price distortion formula for a small tax rate gives the rate of change of the consumer price, $p'(0) = 1/(1 + |-3|/1) = 1/4$. According to the definition of the derivative, the change in the consumer price is equal to $p(t) - p(0) = p(1) - p(0) \approx p'(0)$.

Correct answers: (e)

15. Consider a consumer with the preference relation $u(x_1, x_2, x_3) = \min\{x_1, x_2, x_3\}$ over three goods, where $x_i \geq 0$ holds for all i . She has the initial endowment $e = (1, 2, 3)$. She can buy and sell units of good $i = 1, 2, 3$ at the price p_i . Which price vector from the price vectors (p_1, p_2, p_3) given in the answers puts the consumer in the best position?

- (a) $(2, 1, 3)$
- (b) $(1, 3, 2)$
- (c) $(3, 2, 1)$
- (d) $(1, 2, 3)$
- (e) $(2, 3, 1)$

The consumer will consume a bundle (x_1^*, x_2^*, x_3^*) with $x_1^* = x_2^* = x_3^*$, because otherwise she could improve her position by trading. For the price vector (p_1, p_2, p_3) , the market value of her initial endowment is $Y = 1 + 2p_2 + 3p_3$ and her utility is equal to $Y/(p_1 + p_2 + p_3)$.

Correct answers: (d)

16. A consumer has a monotonic preference relation regarding bundles of two goods. Then the following condition holds true for her preferences:

- (a) $(2, 8) \sim (1, 7)$
- (b) $(2000, 7) \succ (1, 7)$
- (c) $(2000, 7) \prec (1, 7)$
- (d) $(2, 8) \prec (1, 7)$
- (e) $(2, 8) \succ (1, 7)$

Lecture Chapter 1.

Correct answers: (e)

17. The preference relation of a consumer regarding the bundle of goods (x_1, x_2) is quasi-linear, where x_2 represents the money spent on the other goods excluding good 1. The consumer has the willingness-to-pay function $v(x_1) = \ln(x_1 + 1)$. Then the preference relation is represented by the following utility function: $u(x_1, x_2) =$

- (a) x_2 .
- (b) $\ln(x_1 + 1) - x_2$.
- (c) $2 \ln(x_1 + 1) + 2x_2$.
- (d) $1/(x_1 + 1) - x_2$.
- (e) $\ln(x_1 + 1)$.

The money-equivalent form of the utility function is $\ln(x_1 + 1) + x_2$. Multiplying this by 2 is a positive monotonic transformation.

Correct answers: (c)

18. Consider a consumer's demand. The following statement is false for at least one price-income situation:

- (a) The derivative of the demand for good 1 with respect to the price of good 1 is smaller than 0.
- (b) Good 1 is an ordinary good.
- (c) Good 1 is a luxury good.
- (d) Good 1 is inferior.
- (e) Good 1 is normal.

The question here is which statement is false for any given demand for at least one price-income situation. The answer is Good 1 is inferior. Reason: With an income close to 0, consumption can only be close to zero; the quantity consumed cannot always be strictly decreasing with income. Here is the formal argument. Consider any prices. Either the demand is such that no quantity > 0 of good 1 is demanded at any income level. In this case, the quantity demanded is constant and good 1 is therefore not inferior. Or the demand is such that at some income level Y , a quantity $x_1 > 0$ is demanded. Now consider an income level Y' that is so low that $Y' < p_1 x_1$. At this income, the budget constraint necessarily requires a quantity x'_1 to be demanded for which $p_1 x'_1 \leq Y'$ is true, i.e., $x'_1 < x_1$. We thus see that when income increases from Y' to Y , the quantity demanded increases. Therefore, good 1 cannot be inferior everywhere in the income range from Y' to Y .

For all other answers, there are demands for which the statement is true. An example in which good 1 is always a luxury good is $u(x_1, x_2) = \min\{\sqrt{x_1}, x_2\}$. Here, with prices $p_1 = p_2 = 1$, at the optimal bundle (x_1^*, x_2^*) , Walras' law applies, $Y = x_1^* + x_2^*$, and secondly, due to utility maximization, $x_2^* = \sqrt{x_1^*}$, so consumption shifts more and more toward good 1 as income increases.

Correct answers: (d)

19. Consider an Edgeworth exchange economy in which each A -trader has the endowment $(2, 2)$ and each B -trader has the endowment $(0, 3)$. In the competitive equilibrium each B -trader consumes the bundle $(1, 1)$. The B -indifference curve is smooth at the equilibrium point, i.e., the marginal rate of substitution $MRS_{1,2}^B$ is defined there. Then the following applies:

- (a) $MRS_{1,2}^B(1, 1) = -2$.
- (b) $MRS_{1,2}^B(1, 1) = 1/2$.
- (c) $MRS_{1,2}^B(1, 1) = 0$.
- (d) $MRS_{1,2}^B(1, 1) = -1$.
- (e) $MRS_{1,2}^B(1, 1) = -1/2$.

The budget line passes through the endowment point $(0, 3)$ and the selected point $(1, 1)$, so it has a slope of $-(3 - 1)/1 = -2$. Therefore, $GRS_{1,2}^B(1, 1) = -2$ because point $(1, 1)$ is optimal for the B -traders.

Correct answers: (a)

20. Consider an individual in the consumption-leisure model whose preferences are given by the utility function $u(k, l) = k + 2l$, where k is the consumption expenditure and l is the amount of leisure chosen. Let $w > 0$ denote the wage. The individual will not work at all if and only if the wage satisfies the following condition:

- (a) $w > 2$.
- (b) $w = 1$.
- (c) $w \leq 1/2$.
- (d) $w \leq 2$.
- (e) $w > 1/2$.

According to the marginal-utility formula, the indifference curves have a slope of $-(\partial u / \partial k) / (\partial u / \partial l) = -1/2$. The individual will not work if and only if the budget line is at least as steep, i.e., $1/w \geq 1/2$, i.e., $w \leq 2$.

Correct answers: (d)

21. Consider a quasi-linear economy. The market demand is given by $D(p) = \max\{0, 10 - p\}$. There are 9 firms $j = 1, \dots, 9$, each with supply function $S^j(p) = p$, and 10 additional firms $j = 10, \dots, 19$, each with supply function $S^j(p) = 2p$. How many units of the good are traded in equilibrium?

- (a) 0
- (b) $39/3$
- (c) a quantity different from those given in the other answers
- (d) $19/3$
- (e) $9/3$

The market supply is $S(p) = 9p + 10 \cdot 2p = 29p$. The equilibrium condition is $10 - p^* = 29p^*$, i.e. $1/3 = p^*$. The equilibrium quantity is $D(p^*) = 10 - 1/3 = 29/3$.

Correct answers: (c)

22. A consumer with the utility function $u(x_1, x_2) = x_1 + 3x_2$ over bundles of goods (x_1, x_2) (with $x_i \geq 0$ for $i = 1, 2$) chooses the bundle $(10, 0)$. Then the following applies to the prices p_1 and p_2 of the two goods:

- (a) $3p_2 \leq p_1$.
- (b) $p_2 \leq 3p_1$.
- (c) None of the inequalities or equations given in the other answers is correct.
- (d) $p_2 \geq 3p_1$.
- (e) $p_2 = 3p_1$.

If she only consumes good 1, the indifference curve is at least as steep as the budget line at point $(10, 0)$, i.e., according to the marginal utility formula $MRS_{1,2}(10, 0) = -1/3 \leq -p_1/p_2$, i.e., $1/3 \geq p_1/p_2$, i.e., $p_2 \geq 3p_1$.

Correct answers: (d)

23. The (competitive) equilibrium of an exchange economy yields

- (a) a Pareto-dominated allocation.
- (b) an allocation in which all traders are strictly better off than with the initial allocation.
- (c) an allocation in which each trader consumes at least as much of each good as in the initial allocation.
- (d) a Pareto-efficient allocation.
- (e) a Pareto-inefficient allocation.

First welfare theorem. It should be noted that trading one good requires selling something in order to buy something else. It may happen that no trader is strictly better off than with the initial allocation; there may be an equilibrium in which everyone consumes the initial allocation because the *MRS* of the two groups of traders are already equal.

Correct answers: (d)

24. A firm has the production function $f(x_1, x_2) = \sqrt{x_1} + 2\sqrt{x_2}$. The input prices are $p_1 = 1$ and $p_2 = 1$. Then the firm has the following cost function:

- (a) $C(q) = q^2/6$
- (b) $C(q) = q^2/7$
- (c) $C(q) = q^2/8$
- (d) $C(q) = q^2/9$
- (e) $C(q) = q^2/5$

According to the marginal-product formula, $GRTS_{2,1}(x_1, x_2) = -\sqrt{x_2}/(2\sqrt{x_1})$. If we approach the vertical coordinate axis (x_2 axis), i.e., $x_1 \rightarrow 0$, the *GRTS* approaches $-\infty$ and approaches 0 as we approach the horizontal axis. Therefore, there is an inner solution here, i.e., the firm will use a strictly positive quantity x_i^* of each of the two inputs i . According to the first-order condition, $GRTS_{2,1}(x_1^*, x_2^*) = -p_1/p_2 = -1$, i.e., $2\sqrt{x_1^*} = \sqrt{x_2^*}$. Given any desired output quantity q , the optimal input bundle lies on the q -isoquant, i.e., $\sqrt{x_1^*} + 2\sqrt{x_2^*} = q$. Therefore, $\sqrt{x_1^*} + 4\sqrt{x_1^*} = q$. Therefore, $\sqrt{x_1^*} = q/5$. Therefore, $\sqrt{x_2^*} = 2q/5$. Therefore, $C(q) = x_1^* + x_2^* = q^2/25 + 4q^2/25 = q^2/5$.

Correct answers: (e)

25. A consumer with a monotonic preference relation over bundles (x_1, x_2) (with $x_1 \geq 0$ and $x_2 \geq 0$) has the initial endowment $e = (1, 1)$. If the price vector (p_1, p_2) changes from $(2, 3)$ to $(6, 9)$, then the following effect occurs:

- (a) Depending on her preferences, the consumer can be strictly worse off or strictly better off.
- (b) Depending on her preferences, the consumer can be strictly worse off or remain as well off as before.
- (c) The consumer is strictly better off.
- (d) The consumer remains as well off as before.
- (e) The consumer is strictly worse off.

All prices have been multiplied by the same factor $k = 3$ so that there is no change in the budget quantity.

Correct answers: (d)

26. The production function of a firm has constant returns to scale. It costs 4 units of money to produce 3 units of output. Then the following applies to the cost function $C(q)$ (for all $q \geq 0$):

- (a) $C(q) = q^2 - 5$
- (b) $C(q) = q + 1$
- (c) $C(q) = 3q/4$
- (d) $C(q) = 4$
- (e) $C(q) = 4q/3$

Due to constant returns to scale, the average costs are constant (see Chapter 7), i.e., $DK(q) = z$ for any $z \geq 0$. Therefore, $C(q) = zq$. Since $C(3) = 4$, it follows that $z3 = 4$, i.e., $z = 4/3$.

Correct answers: (e)

27. A consumer has an indifference curve consisting of the set of points $\{(x_1, x_2) \mid x_1 + 3x_2 = 7, x_1 \geq 0, x_2 \geq 0\}$. At the point $(1, 2)$, what is her good-2-willingness to pay for 6 additional units of good 1?

- (a) 5
- (b) 2
- (c) 3
- (d) 1
- (e) 7

Correct answers: (b)

28. In a quasi-linear economy, consider an individual i with the marginal willingness-to-pay function $(v^i)'(x^i) = e^{-x^i}$ for all $x^i \geq 0$, where e^z (for any real number z) is the exponential function (i.e., the derivative with respect to z is again e^z). The individual consumes $x^i = 2$ units of the good. Denote by $\hat{p}$ the price that is fixed by the government. In this situation, we say that the individual is rationed if and only if

- (a) $\hat{p} \geq e^{-2}$.
- (b) $\hat{p} \leq e^{-2}$.
- (c) $\hat{p} = e^{-2}$.
- (d) $\hat{p} > e^{-2}$.
- (e) $\hat{p} < e^{-2}$.

Rationing is defined as $x^i < D^i(\hat{p})$, i.e. $(v^i)'(x^i) > \hat{p}$. See Chapter 10.

Correct answers: (e)

29. A consumer with a monotonic preference relation over bundles of two goods has the following demand for good 1: $d_1(p_1, p_2, Y) = Y/(10p_1)$. What is her demand for good 2?

- (a) $d_2(p_1, p_2, Y) = 9Y/(10p_1)$
- (b) $d_2(p_1, p_2, Y) = 9Y/(10p_2)$
- (c) $d_2(p_1, p_2, Y) = 0$
- (d) $d_2(p_1, p_2, Y) = Y/(9p_1)$
- (e) $d_2(p_1, p_2, Y) = Y/(9p_2)$

Apply Walras' law.

Correct answers: (b)

30. A bakery has purchased an oven (capital good) at the beginning of the year 2024 for EUR 30,000. The interest rate for financial investments that year is 2%. The total cost of using the oven in 2024 (billed at the end of the year) was EUR 2600. The depreciation rate was therefore

- (a) $1/16$.
- (b) $1/20$.
- (c) $1/19$.
- (d) $1/18$.
- (e) $1/15$.

EUR 2600 = purchase price $(1+0.02)$ - sale price = 30,600 - sale price. So sale price = 28,000, therefore depreciation rate = $(\text{purchase price} - \text{sale price}) / \text{purchase price} = 2,000 / 30,000 = 1/15$

Correct answers: (e)